

Data Management Project Proposal

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Project Type

[DW] Select a set of data sources, integrate them by means of ETL operations (you can either use an integration tool or write simple scripts to solve common ETL problems), define the DFM model for your data, define and populate the corresponding star schema or snowflake schema, by using a relational DB (e.g., Postgres). Perform OLAP queries.

Focus: ability to combine data from different datasets in order to produce a relevant unified source of analysis.

Datasets

I'm considering the following datasets:

1. **Woodland Trust Climate Change Phenology Dataset**
Dataset containing phenological events (flowering, first leaf etc) of multiple species around the UK.
<https://www.kaggle.com/datasets/thomaswrightanderson/woodland-trust-climate-change-phenology>
2. **Meteostat Weather Dataset**
Global weather and meteorological dataset.
<https://meteostat.net/en/>

Project Description

The goal of the project is to study how climate variables influence phenological events over time. I will design and implement an ETL pipeline in python to clean and integrate the datasets. The data will be modeled using a dimensional approach (DFM) and implemented as a star schema in PostgreSQL.

The warehouse will include fact tables containing phenological observations and weather data, along with dimension tables such as time, location, species, and phenological event type. I will perform OLAP queries to analyze trends across years, species, and regions. Example analyses include detecting changes in flowering periods over time and evaluating correlations between temperature increases and shifts in seasonal biological events.